Code Drop 1

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The current goal for code drop #1 was to get a 3D CPU verison of a ray marcher running that can display the Mandelbulb fractal, along with a project setup via CMake and OpenGL that we can build on. All of that is in place.

The project builds cleanly with CMake, it has separate Debug and Release configurations. GLFW and OpenGL are dynamically linked. GLFW handles window creation, the OpenGL context, and user input, while OpenGL itself (via Mesa, the open-source implementation on Linux) handles rendering. GLAD is used to load the correct OpenGL function pointers at runtime. GLM provides the math primitives (vectors, matrices) that we may use. Dear ImGui is compiled as its own static library target and ready to hook up to get interactive controls.

So far GLFW opens a window and sets up the OpenGL context, then the renderer and runs a ray marcher that sphere traces the Mandelbulb's distance field, with basic shading to show the 3D surface. It's single-threaded on the CPU for now, which was the goal for this stage since figuring out how to get OpenGL and the other libraries working together was a significant amount of work.

We also have some of the interactivity set up that we planned to have for the 4/15 code drop #2.

For the 4/15 drop, the plan is to move the ray marcher to the GPU so rays are traced in parallel, and to add keep/extend UI and interactivity using ImGui.

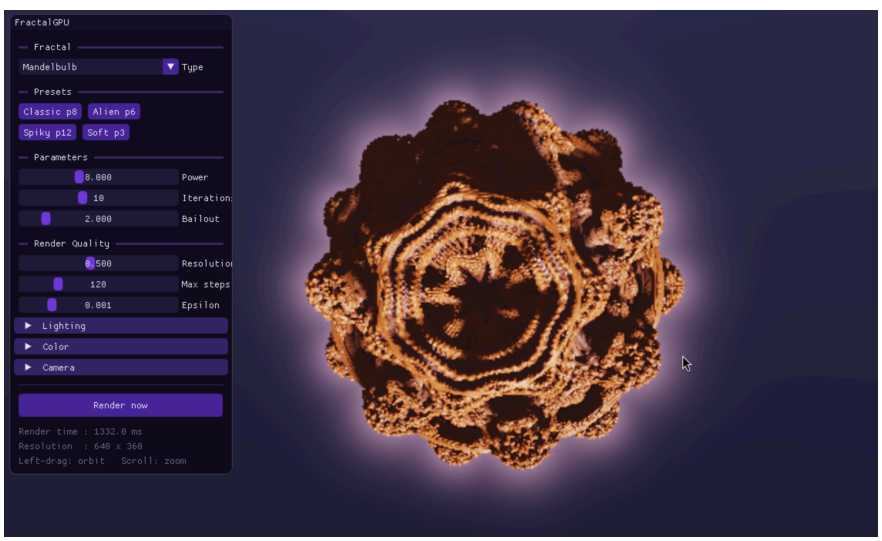

Figure 1: Rendered Mandelbulb fractal on CPU with GUI.
(You can see the timings are very slow on CPU, 1332ms per frame!)